

Evaluating the Potential Impacts of WA DNR Recreation Closures

DNR Project Group

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Executive Summary

This project was undertaken to explore the impacts of site closures and service reductions in the Washington Department of Natural Resources (DNR), following their 2026 budget reduction. DNR manages outdoor recreation sites across the state, providing recreational opportunities to millions of visitors. However, following these closures, there is a knowledge gap in how the specific distribution of these closures may impact local communities differently, requiring a close analysis of potential demographics and environmental disparities. This corroborates DNR's mission of providing equitable access and service to all residents across Washington State.

To address these questions, our team analyzed recreation site data from Washington State DNR, the U.S. Forest Service (USFS), and Washington State Parks & Recreation alongside demographic and environmental health datasets, provided to us by the project sponsor at WA DNR. The team evaluated how nearby recreation opportunities changed before and after the proposed closures and assessed the extent to which recreation opportunities from other agencies could serve as substitutes for lost DNR sites. Furthermore, we overlaid this data with information on demographics and environmental health to deepen our analysis of any potential disproportionate demographic impacts following the site closures.

Our analysis found that the impacts of the proposed closures varied by recreation activity. This project analyzed the five most reduced activities following DNR changes, which, in order, are: hiking, camping, scenic viewing, fishing, and kayaking. We also analyzed the distribution of alternative sites providing the same activities lost after site closures within a 10-mile radius. Hiking experienced the largest numerical reduction, but had sufficient alternative sites for each closed location. Scenic opportunities also declined, although impacts were concentrated within a smaller number of census tracts, which all had available alternative sites. In contrast, camping opportunities showed little measurable change after closures when recreation opportunities from multiple agencies were considered, minimizing the negative impacts of closing camping sites. Fishing and kayaking only had a few site closures, but had some sites without nearby alternatives. Bear Creek Campground, Rock Lakes Campground, and Chopaka Lakes Campground all do not have a nearby alternative site within a 10-mile radius providing the same activity lost during site closures. Otherwise, all other sites and activities analyzed in this project have at least one alternative site in the specified radius.

The analysis also found that some communities continued to lack nearby recreation opportunities even after accounting for recreation sites from DNR, USFS, and Washington State Parks, suggesting that gaps in recreation opportunity distribution existed before the closures and remained afterward. Demographic analyses indicated that these gaps were often associated with communities continuing higher proportions of non-white and Hispanic residents, highlighting the importance of considering equity alongside recreation planning decisions.

Project outcomes included an ArcGIS Online web map, activity-specific recreation opportunity layers, demographic and environmental equity analyses, summary tables, a final GIS report, and a data package to deliver to Washington DNR. These products provide DNR with a data-driven assessment of potential community impacts and can support future recreation planning, accessibility evaluations, and resource allocation decisions.

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1. Sponsor, Background, and Problem Statement

This section will summarize the sponsor's role and activities, the challenges currently faced by the project sponsor, and identify the problems this project will address. It will include background information on the budget crisis that led to this project's formation and the broader necessity of this project.

The sponsor of this project is the Washington State Department of Natural Resources ([DNR](#)). They are tasked with sustaining and protecting the health and productivity of Washington State's natural land and water resources. As part of this role, DNR is responsible for providing recreational opportunities for over 200 recreation sites in Washington State, covering 1,300 miles of trails and 3.1 million acres of land. Other responsibilities include conserving environmental and cultural resources, improving equitable access across its sites, planning for and mitigating wildfires, and cultivating environmental stewardship.

In 2026, the [Washington State Governor's office reduced funding](#) for the DNR's Recreation Program. This exacerbated an existing backlog of deferred maintenance across its sites. Because of this funding shortfall, DNR was forced to close or reduce service for 25 sites, with no estimate for when normal service will resume ([McEvoy, 2026](#)).

In light of these consequential changes, the Department of Natural Resources has reached out to the Geography Department at the University of Washington for assistance on a project to analyze the greater impact of these site closures and service reductions. This project was formed in response. We aim to answer these questions using GIS tools and applications, highlighting disparate demographic or area impacts across Washington State, and addressing the problem of the undefined impacts of closing and reducing service for 25 sites across Washington State.

2. Data Analysis Strategy

Our first tasks with this assignment were preparing our datasets and identifying the roles needed for this project. We received an initial dataset including all of the DNR recreation sites. This was edited down to include only the ones that were (at the time) being considered for closure. When we received confirmation that the final list of sites had been chosen, we amended our DNR site data to reflect that. We additionally received US Forest Service data (USFS), which was also cleaned and edited to contain only the recreation sites within Washington State. We repeated this process with our Washington State Parks data as well. The USFS and State Parks data were primarily used to identify suitable replacement sites for the DNR sites that would be closed. We also received data connected to equity and accessibility, which was cleaned and used to visualize the comparison between pre - and - post site closures. Once we completed all of the data cleaning and preparation, we visualized the data in ArcGIS Pro. For the replacement sites, we split the data up into categories of activity and displayed all of the open sites relevant to each category. Filtering by category allowed us to analyze each one individually and the specific sites it impacts. A similar process was used to create our accessibility maps using a location buffer to measure nearby rec accessibility opportunities. After this stage, we shared our web layers to ArcGIS Online so they can all be visually represented in an interactive way.

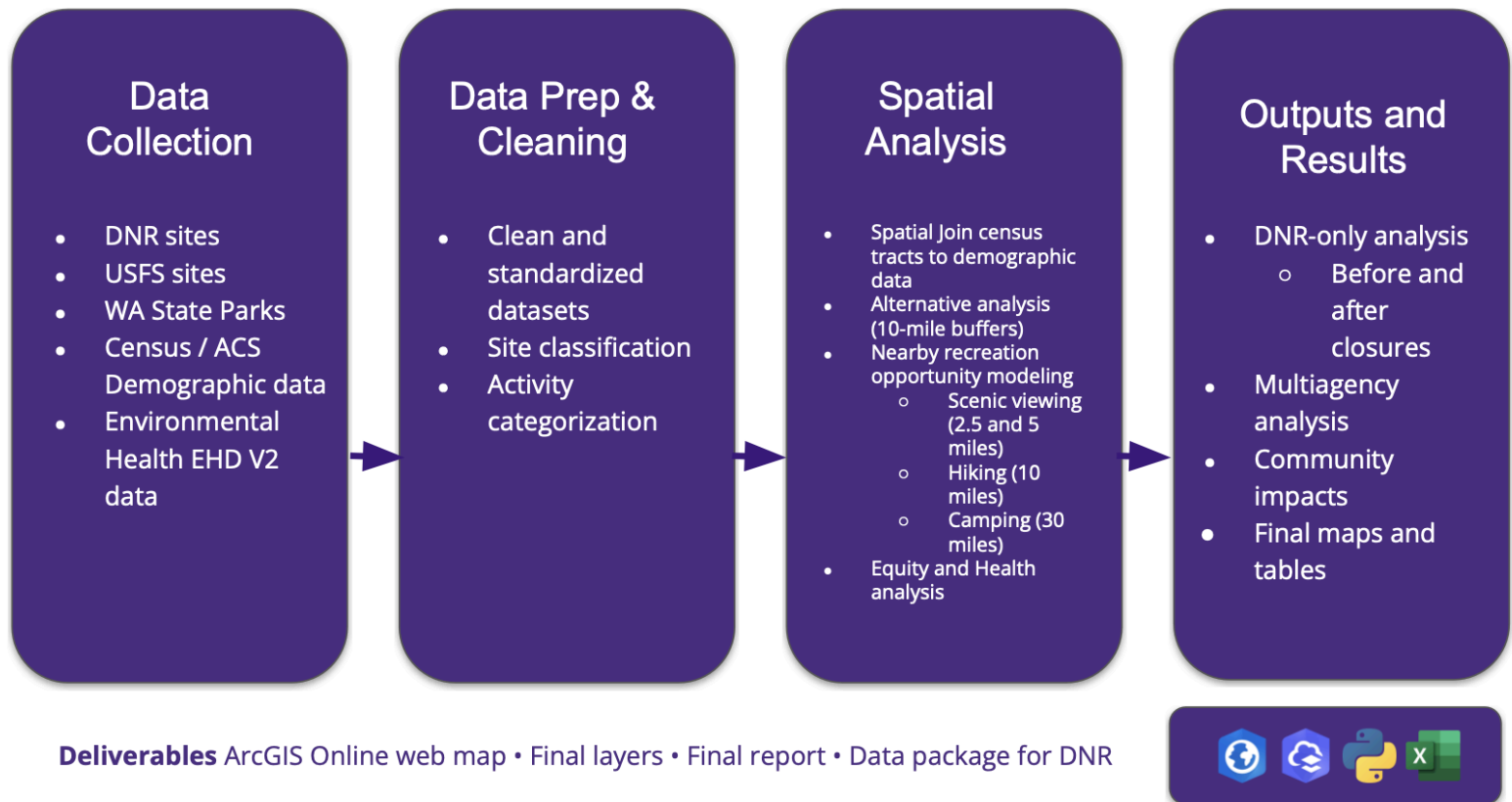


Figure 1. Project workflow used to prepare data, model alternative and nearby recreation opportunities, conduct demographic and equity analyses, and generate final project deliverables.

3. Project Digital Environments

This section describes the digital environments and collaborative platforms used to support the project workflow. It outlines the GIS software, cloud-based systems, and communication tools used to organize data, coordinate project tasks, and develop project deliverables. This section also explains how geospatial data was processed, shared, and visualized across different platforms throughout the analysis.

3.1 GIS and Mapping Software

ArcGIS Pro was the primary software application used for this project. It was the main environment for all spatial data processing, geoprocessing workflows, and map production. ArcGIS Pro was used to clean and organize datasets, perform spatial joins, generate proximity buffers, run summary statistics, and produce choropleth maps and activity-specific recreation layers. The software supported the team's modeled nearby recreation opportunity analysis, demographic comparisons, environmental equity mapping, and alternative recreation site analysis.

ArcGIS Online was used to host, organize, and share project outputs for both the course and the project sponsor. Final layers were published from ArcGIS Pro to ArcGIS Online through the University of Washington's organizational account, allowing team members, professors, and the sponsor to access the same datasets and map products.

Python scripting supported several data preparation workflows used across the project.

Recreational activities in the DNR and USFS datasets were originally stored as comma-separated values within a single field, which made activity-specific analysis difficult. To fix this, a Python script was used to split and expand those activities into separate binary columns coded as 0 or 1. This made it easier to isolate and analyze activities such as hiking, camping, fishing, kayaking, and scenic viewing during later stages of the project.

Microsoft Excel was used as a supplementary tool for organizing and reviewing summary statistics exported from ArcGIS Pro. Demographic summaries, tract counts, and before-and-after comparisons of modeled nearby recreation opportunities were formatted in the program to support data interpretation, presentation preparation, and report writing.

3.2 Collaboration and Shared Storage

Google Drive served as the team's primary shared storage environment for report drafts, presentation materials, and datasets. A shared folder ensured that team members had access to the most current versions of deliverables throughout the quarter.

Discord was used as the team's primary communication platform throughout the project. Channels were used for general discussion, shared datasets, screenshots, workflow updates, troubleshooting, clarification, and quick project revisions. The platform was especially useful for

coordinating responsibilities between milestones and discussing workflows and analysis decisions as the project evolved.

The University of Washington ArcGIS Online organizational account also acted as a collaborative workspace where hosted feature layers, web maps, and spatial outputs could be shared between team members and the project sponsor.

3.3 Final Deliverable

Final project deliverables include a shared ArcGIS Online web map containing our final layers. The web map is intended to support interactive visualization. In addition to the web map, we will provide exported map layers and supporting geospatial data as compressed ZIP files for sponsor use outside the UW ArcGIS organization. Additional supporting materials, including maps, summary tables, and final report documentation, will also be provided for sponsor review and future project use.

4. Data Acquisition

This section covers the sources of all the data that we used, and the reasoning behind using them. We will also discuss any data quality issues that we came across during this project. This section aims to help readers understand where the data comes from and to ensure reproducibility in future replications and research.

The sources of the data are as follows:

DNR Sites — Washington Spatial Data

USFS Recreational Opportunities — U.S. Forest Service - Geospatial Data Discovery

WA State Parks and Recreation Campsites — Washington Spatial Data

Environmental Health Disparities Data — Washington Spatial Data

U.S. Census Bureau American Community Survey (ACS) 5-Year Estimates

TIGER/Line Census Tract Shapefiles — U.S. Census Bureau

4.1 Recreation Site Data

DNR Site data contained point data on all the DNR sites in Washington State. USFS Recreational Opportunities data contained point data for all USFS locations in the US. WA State Parks and Recreation Campsites showed all campsites managed by WA State Parks and Recreation as point data. We had to clean the data in a few steps: first, we truncated the USFS data to only contain site data within Washington State; then, we manually coded DNR sites as open or closed based on the information given to us by the project sponsor; lastly, because information on activities was all stored in a single column for DNR and USFS data, with each activity split by a comma, we utilized Python to split the activities so that each activity had its own column in their corresponding dataset. All of the sites would be coded as 0 or 1 for each of the activity columns, helping us identify and isolate sites by activity.

4.2 ACS Demographic Data

To support the demographic and equity analysis components of the project, five-year estimates from the U.S. Census Bureau's American Community Survey (ACS) were joined to Washington State census tract boundaries using TIGER/Line shapefiles. The ACS data provided tract-level demographic and socioeconomic attributes, including median household income, total population, racial composition (percent non-white, percent Hispanic, percent Black, percent Asian), disability rates, households without access to a vehicle, and the proportion of elderly residents aged 65 and over. Census tract boundaries served as the primary spatial unit for our proximity-based modeled recreation opportunity analysis, and demographic tables were joined to tract shapefiles using GEOID field. A total of 1,784 census tracts representing all census tracts within Washington State were included in the project.

4.3 Environmental Health Disparities Data

Our project included an analysis of environmental and health considerations in relation to WA recreation sites. We sought ways to study the effects of recreation site service changes on different populations, now and in the future. Information about visitor demographics can be difficult to find without further investigation. We chose to use publicly available census tract data to observe the different burdens faced by local communities. We found this with The Washington Environmental Health Disparities Map version 2, created in 2022 by the Washington Department of Health. Full documentation for the dataset, including the attribute definitions, is included [here](#). This dataset was joined to our DNR and USFS point data, so that each site contained the Environmental Health Disparities data from its parent census tract.

The Environmental Health Disparities Map (EHD / EHD V2) uses a set of 19 indicators for health burdens. Each indicator represents an environmental or socioeconomic source of hazard, suffering, or disadvantage experienced by a population at the census tract level. We excluded the raw data and used only the Index rankings created by the dataset authors. Indicators are ranked from 1 – 10. A ranking of 1 means 90% of other census tracts experience a higher burden from this indicator (in other words, the current census tract experiences the lowest level of burden from this indicator). A ranking of 9 means that only 10% of the other census tracts experience a higher level of burden from this indicator.

These 19 individual indicators are grouped into four categories: Environmental Exposure; Environmental Effects; Socioeconomic Factors; and Sensitive Populations. The four categories are then each given an Index rank equivalent to the average rank of their indicators. Finally, the EHD V2 ranking uses a weighted combination of these four categories to represent the overall level of burden experienced by each census tract.

5. Findings and Discussion of Results

This section will discuss the findings and outputs of the project using the created products and visualizations. We will discuss which demographics are most affected by the site closures and service reductions, and analyze which activities experienced the greatest reductions in count and are the least replaceable. We will also provide supporting evidence using supplementary findings from our project, including identifying alternative sites for lost activities.

5.1 Alternative Analysis

Beginning with site alternative analysis, we find that most closed DNR sites have adequate alternative sites within a 10-mile radius. We will go into further detail about the exceptions. Due to the large number of activities across DNR sites, we have opted to only focus on the five activities that saw the greatest losses following the site impacts, which are camping, hiking, fishing, kayaking, and scenic viewing. Firstly, for our camping activity analysis, all closed DNR sites have at least one alternative camping site through either other open DNR sites, USFS sites, or WA State Park sites. WA State Park data only included campgrounds, so subsequent activity analysis will not use this data and will only use site data from USFS and DNR.

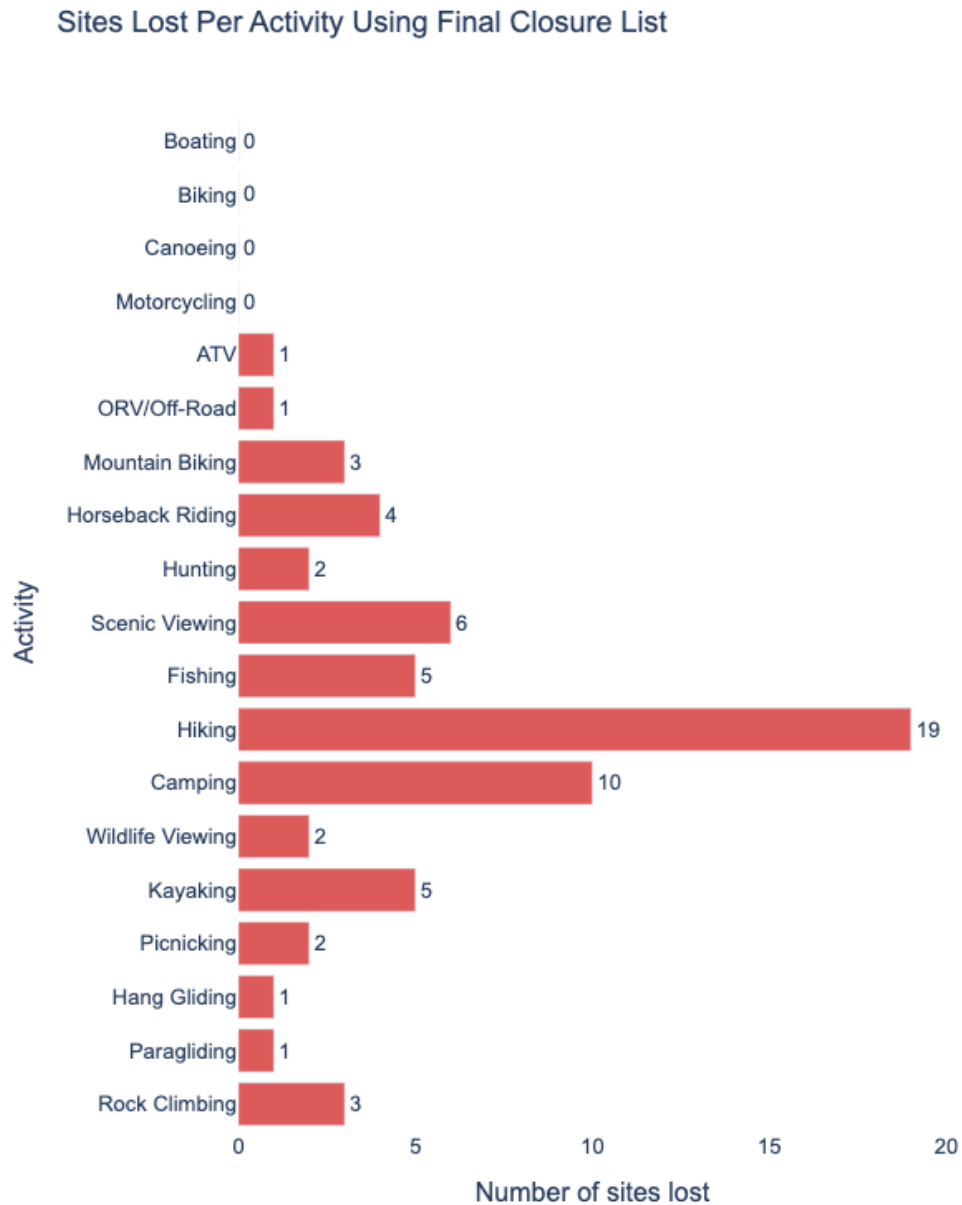


Figure 2. Bar graph of all the reductions in recreational activities following the closures and service impacts of DNR Sites. Note that Snoqualmie sites are included in this graph and the subsequent visualizations.

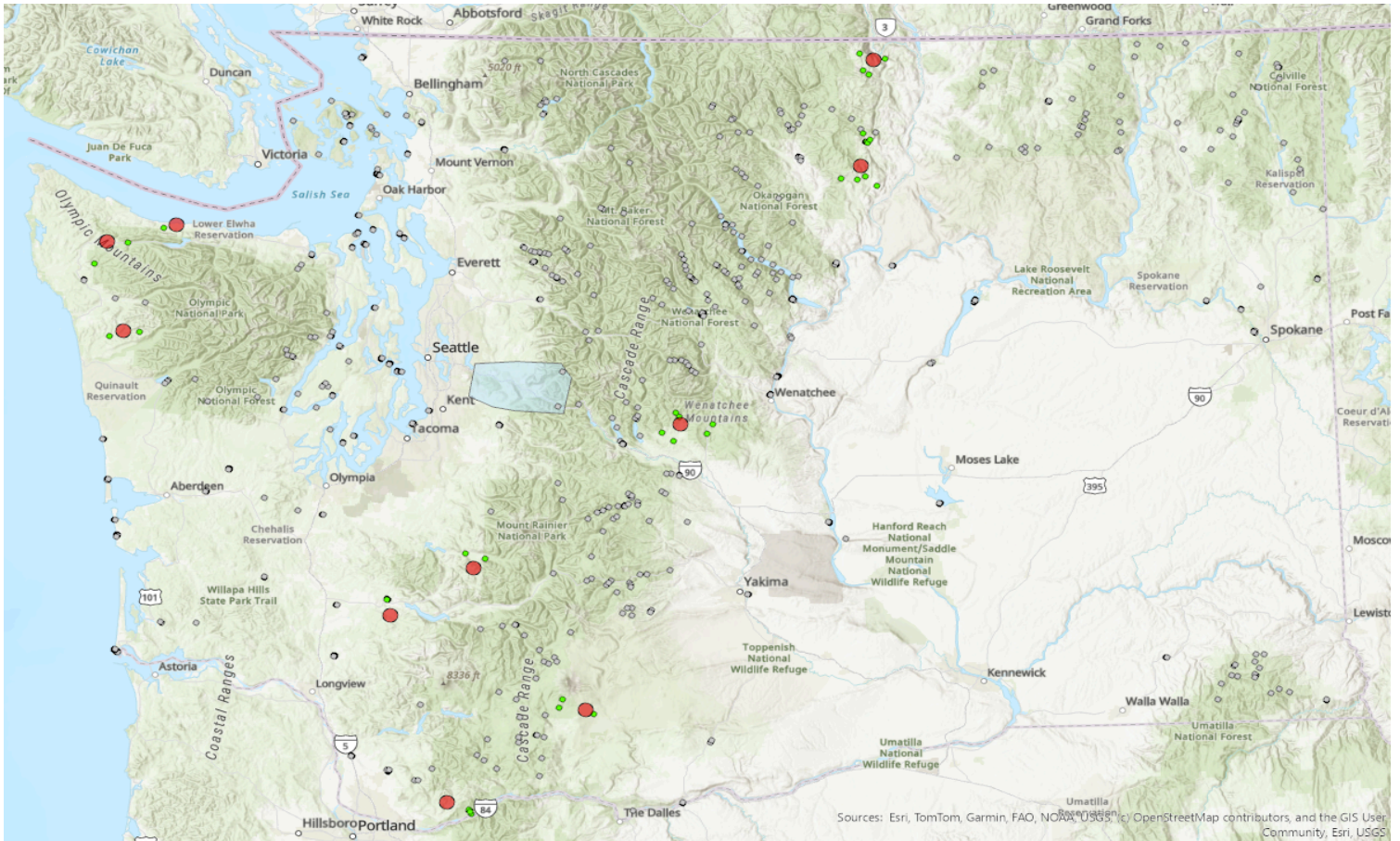


Figure 3. Alternative site analysis for camping activities. Red dots symbolize closing DNR sites. Green dots indicate open alternative sites within a 10 mile radius, including open DNR sites, USFS sites, and WA State Parks. Gray sites show open sites with the same activity outside of the 10 mile radius. The blue polygon represents the Snoqualmie sites, which experience seasonal bathroom closures instead of site closures. The color-coding holds for all subsequent visualizations.

Our second analysis is for hiking activities. Note that Bear Creek Campground does not have an alternative site providing the same activity within a 10 mile radius. Additionally, no alternatives exist for the Snoqualmie sites, requiring users to attend the sites with possible bathroom closures. Out of the four closed sites, only three have nearby alternatives.

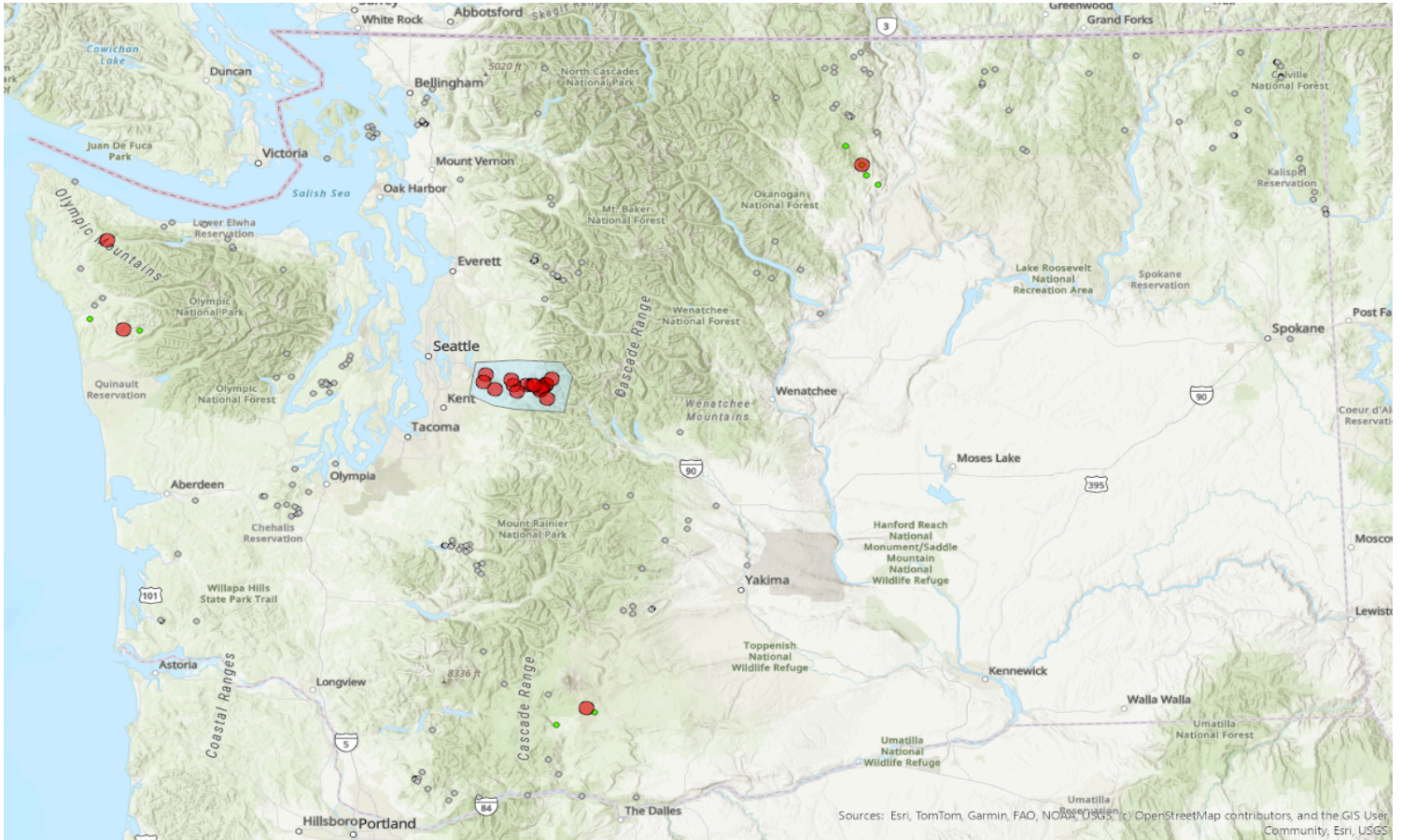


Figure 4. Alternative site analysis for Hiking.

For fishing activities, Bear Creek Campground and Rock Lakes Campground do not have alternative sites within a 10-mile radius. Out of the five closed DNR sites, only three have nearby alternatives.

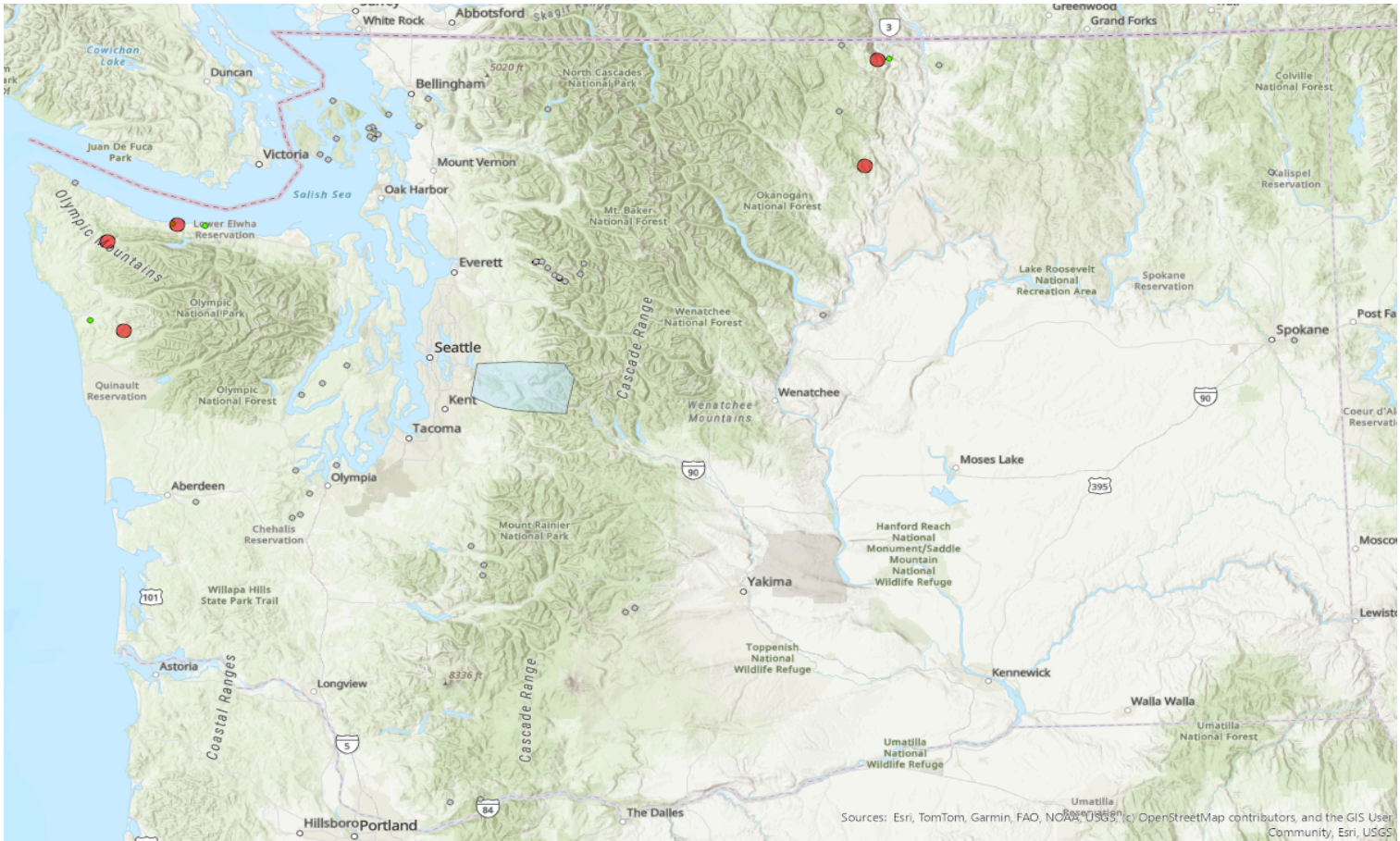


Figure 5. Alternative site analysis for Fishing.

For Kayaking, none of the Snoqualmie sites have a nearby alternative, and the Chopaka Lake Campground does not have an alternative either. Out of the two closed sites, one does not have a nearby alternative.

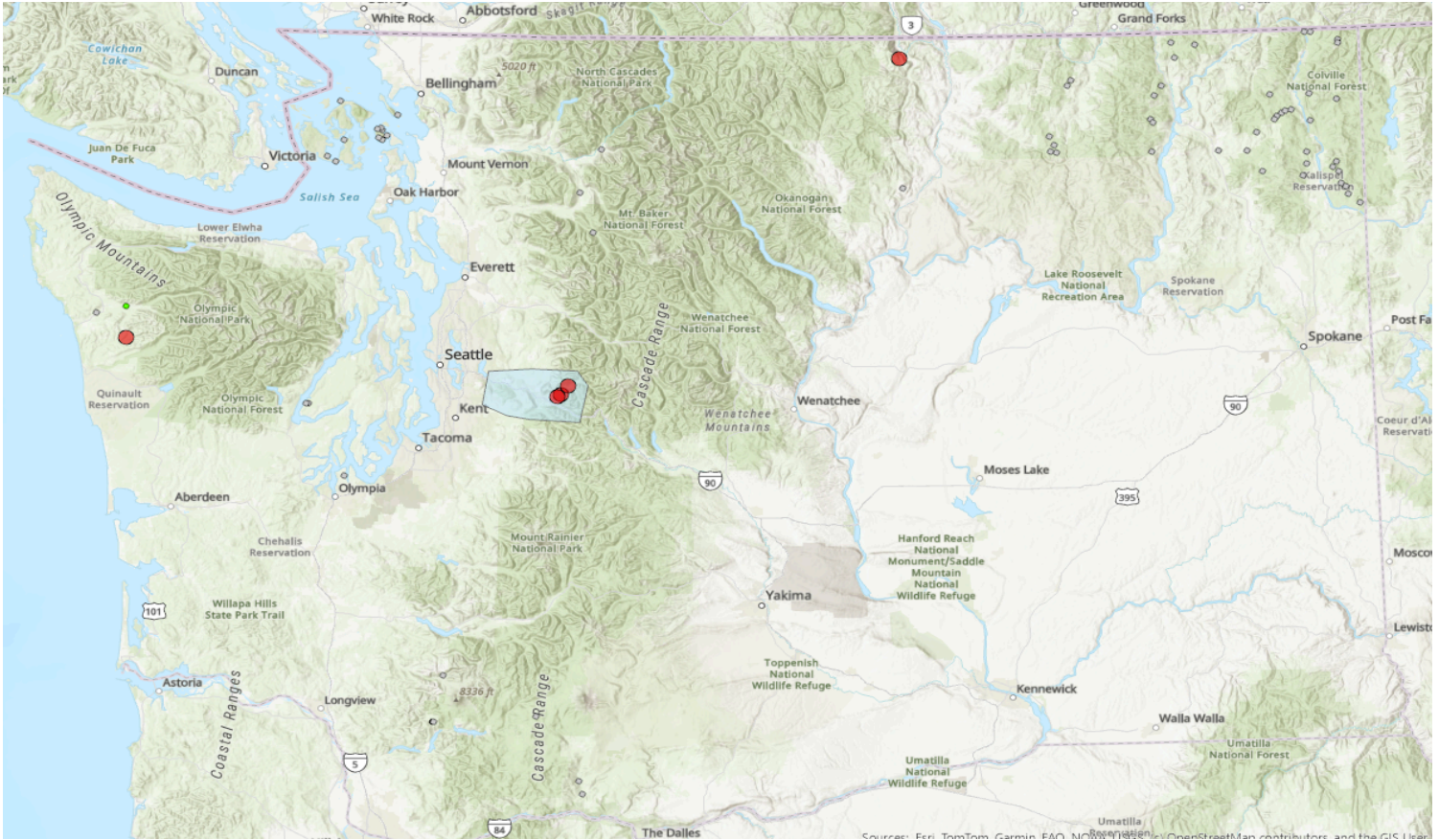


Figure 6. Alternative site analysis for Kayaking.

For scenic viewing activities, most of the affected sites are in the Snoqualmie area, which will experience seasonal bathroom closures. The only non-Snoqualmie site being closed is Dougan Creek Campground, which has a nearby alternative site.

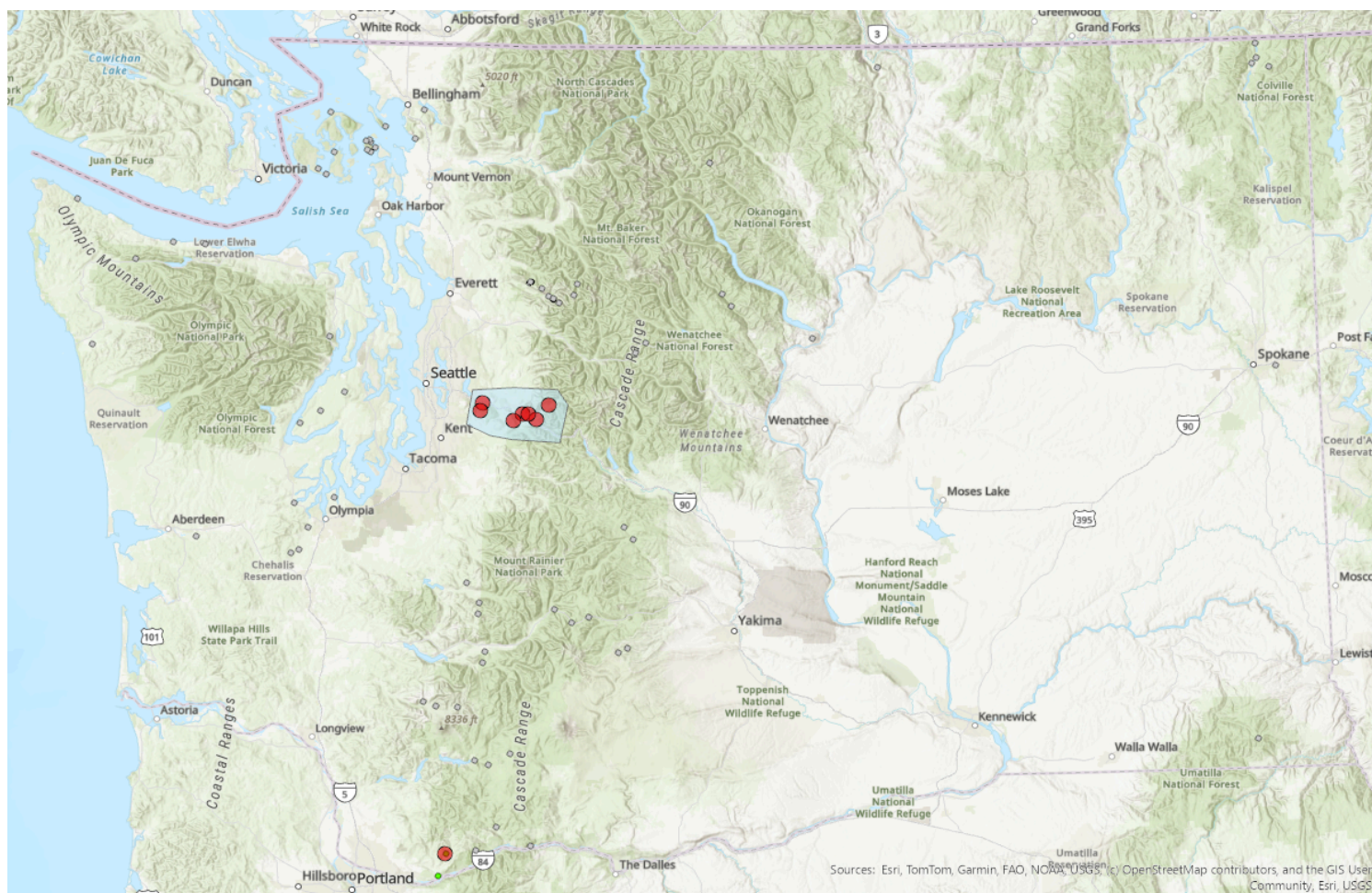


Figure 7. Alternative site analysis for Scenic Viewing.

Our site alternative analysis has provided us with some significant findings; firstly, a vast majority of closed sites have an alternative open site within a 10 mile radius providing the same activity. There are only three closed sites that do not possess an alternative, keeping activity impacts at a minimum across Washington State. However, none of the Snoqualmie sites had a suitable alternative for any of its activities, which may require users to experience discomfort or inconvenience with seasonal bathroom closures.

5.2 Proximity-Based Modeled Recreation Opportunity Framework

To evaluate the potential impacts of proposed DNR recreation closures on communities across Washington State, this analysis modeled nearby recreation opportunities at the census tract level. Rather than measuring accessibility from the perspective of individual recreation sites, the framework focused on whether census tracts had at least one nearby recreation opportunity within a specified distance threshold ($\text{Join_Count} > 0$) or no nearby opportunities within that threshold ($\text{Join_Count} = 0$). Distance thresholds were selected to reflect different recreation travel patterns. Scenic viewing was analyzed at 2.5 and 5-mile thresholds because it is generally considered a shorter-distance and more opportunistic activity. Hiking was analyzed at 10 miles to

reflect its role as a planned day-trip, while camping was analyzed at 30 miles since camping destinations involve longer travel distances.

The findings should be interpreted as proximity-based measures of nearby recreation opportunities rather than guaranteed recreation access or actual recreation behavior. Transportation availability, physical ability, financial barriers, and recreation preferences all influence participation in ways that are not fully captured through proximity-based modeling alone.

5.2.1 Overall Recreation Opportunity Changes Before and After Closures

The overall accessibility analysis compared nearby recreation opportunities before and after DNR closures across scenic viewing, hiking, and camping activities. Results varied depending on recreation type and the availability of substitute opportunities from other agencies.

Proximity-based modeled accessibility. Unit of analysis: census tracts. Total WA tracts: 1,784

SCENIC VIEWING	Distance	Tracts with nearby opportunities before	Tracts with nearby opportunities after	Change
Scenic viewing	2.5 mi	116	101	-15
Scenic viewing	5 mi	237	201	-36

HIKING	Distance	Tracts with nearby opportunities before	Tracts with nearby opportunities after	Change
Hiking (dnr only)	10mi	637	513	-124
Hiking (multiagency)	10 mi	658	536	-122

CAMPING	Distance	Tracts with nearby opportunities before	Tracts with nearby opportunities after	Change
Camping (dnr only)	30 mi	1,523	1,519	-4
Camping (multiagency)	30 mi	1,714	1,714	0

Table 1. Recreation accessibility summary table before and after DNR closures. Tracts with nearby modeled recreation opportunities = join count > 0. Distance thresholds reflect activity-specific travel behavior: scenic viewing (2.5 - 5 mi), hiking (10 mi), camping (30 mi). Multiagency = DNR + USFS + Washington State Parks.

Table 1 shows that hiking experienced the largest reductions across both DNR-only and multiagency scenarios. The number of tracts with nearby DNR-only hiking opportunities decreased by 124 tracts, while the multiagency scenario decreased by 122 tracts after closures. Scenic viewing also showed measurable reductions at both the 2.5-mile and 5-mile thresholds,

with 15 and 36 fewer tracts. Camping accessibility remained relatively stable even after closures were applied. This suggests that substitution effects varied by recreation activity. Hiking opportunities appeared much more difficult to replace locally, while camping opportunities remained more geographically distributed across DNR, USFS, and Washington State Park sites.

5.2.2 Baseline DNR Demographic and Equity Analysis

To examine the equity implications of the closures, demographic characteristics were compared between census tracts with and without nearby DNR recreation sites before and after closures. This analysis included all DNR recreation sites using a 10-mile threshold. Before closures, 680 census tracts had at least one nearby DNR recreation site within 10 miles. After closures, that number dropped to 553 tracts, meaning 127 tracts lost nearby DNR recreation opportunities.

Tracts with nearby modeled DNR recreation opportunities (Join Count > 0) - All activities

	Before 680 tracts	After 553 tracts	Change
Median Household Income	\$101,167	\$87,371	-\$13,796
Population	2,847,128	2,246,353	-600,775
% non-white	30.3%	26.2%	-4.1%
% Hispanic	12.0%	12.7%	0.7%
% Black	2.1%	1.6%	-0.5%
% Asian	7.4%	2.9%	-4.5%
% with disability	14.7%	16.0%	1.3%
% no vehicle	5.0%	5.0%	0.0%
% elderly	20.5%	20.5%	0.0%

Tracts with no nearby modeled DNR recreation opportunities (Join Count = 0) - All activities

	Before 1,104 tracts	After 1,231 tracts	Change
Median Household Income	\$106,722	\$112,052	\$5,330
Population	4,968,988	5,569,763	600,775
% non-white	39.4%	40.3%	0.9%
% Hispanic	15.5%	14.8%	-0.7%
% Black	4.6%	4.6%	0.0%
% Asian	10.2%	11.9%	1.7%
% with disability	13.5%	13.1%	-0.4%
% no vehicle	7.5%	7.2%	-0.3%
% elderly	18.5%	18.5%	0.0%

Table 2. Demographic comparison of census tracts with and without nearby modeled DNR recreation opportunities before and after proposed closures.

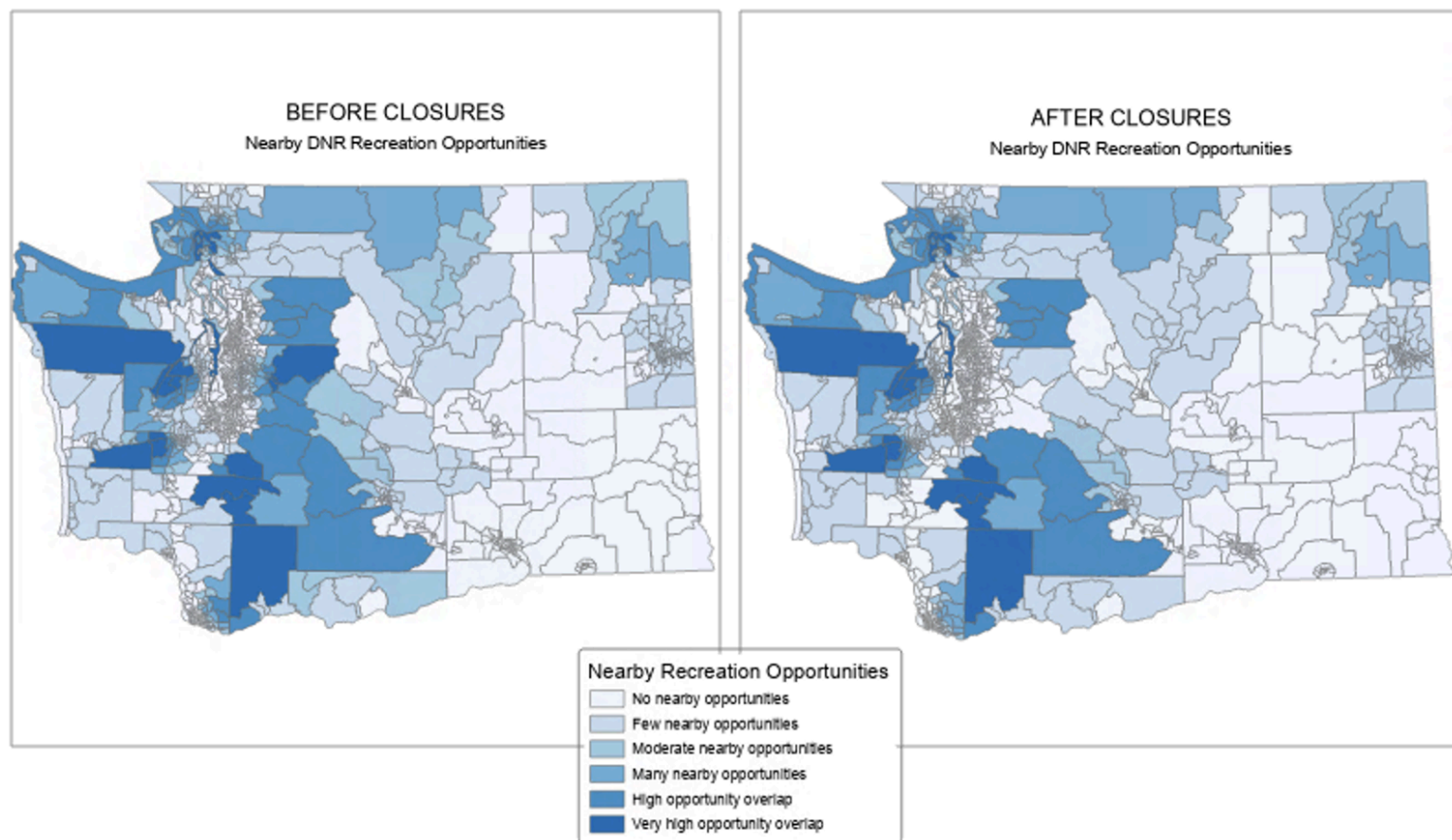


Figure 8. Nearby DNR Recreation Opportunities before and after site closures. Census tracts classified by the number of nearby DNR recreation opportunities within the modeled threshold.

Table 2 and Figure 8 show that nearby recreation opportunities declined across parts of western and central Washington following the closures. Tracts that retained nearby recreation opportunities had lower median household incomes after closures, decreasing from \$101,167 to \$87,371, suggesting that many higher-income communities were part of those that lost nearby DNR recreation sites. The percentage of non-white residents also declined from 30.3% to 26.2%, with the largest shift occurring among Asian populations, which decreased from 7.4% to 2.9%. Disability rates increased slightly from 14.7% to 16.0%, while no-vehicle and elderly population remained stable. The findings suggest that recreation opportunities were unevenly distributed across communities before closures and became somewhat more uneven afterwards.

5.2.3 Activity-Specific Recreation Opportunities - Hiking

Hiking experienced the largest reduction in nearby recreation opportunities and was analyzed separately using a 10-mile threshold. The analysis compared DNR-only hiking accessibility with a broader multiagency scenario that included remaining open DNR and USFS hiking sites.

Tracts with nearby modeled hiking opportunities: accessible (join count > 0)

	DNR only 637 → 513			Multiagency 658 → 536		
	Before	After	Change	Before	After	Change
Median Household Income	\$100,406	\$85,777	-\$14,629	\$99,480	\$85,444	-\$14,036
Population	2,671,781	2,084,528	-587,253	2,729,659	2,150,419	-579,240
% non-white	30.5%	26.1%	-4.4%	30.2%	25.8%	-4.4%
% Hispanic	11.9%	12.6%	0.7%	11.6%	12.3%	0.7%
% Black	2.1%	1.6%	-0.5%	2.1%	1.5%	-0.6%
% Asian	7.6%	2.8%	-4.8%	7.4%	2.8%	-4.6%
% with disability	14.7%	16.1%	1.4%	14.9%	16.3%	1.4%
% no vehicle	5.1%	5.2%	0.1%	5.0%	5.1%	0.1%
% elderly	20.9%	20.9%	0.0%	21.6%	21.6%	0.0%

Table 3. Demographic comparison of census tracts with nearby modeled hiking opportunities before and after DNR closures. DNR-only analysis includes only DNR hiking sites. Multiagency analysis includes remaining open DNR and USFS hiking sites within a 10-mile threshold.

The number of tracts with nearby hiking opportunities declined from 637 to 513 in the DNR-only analysis and from 658 to 536 in the multiagency analysis. Because the two scenarios differed by only two tracts after closures, the results suggest that USFS provided very limited local substitution for lost DNR hiking opportunities.

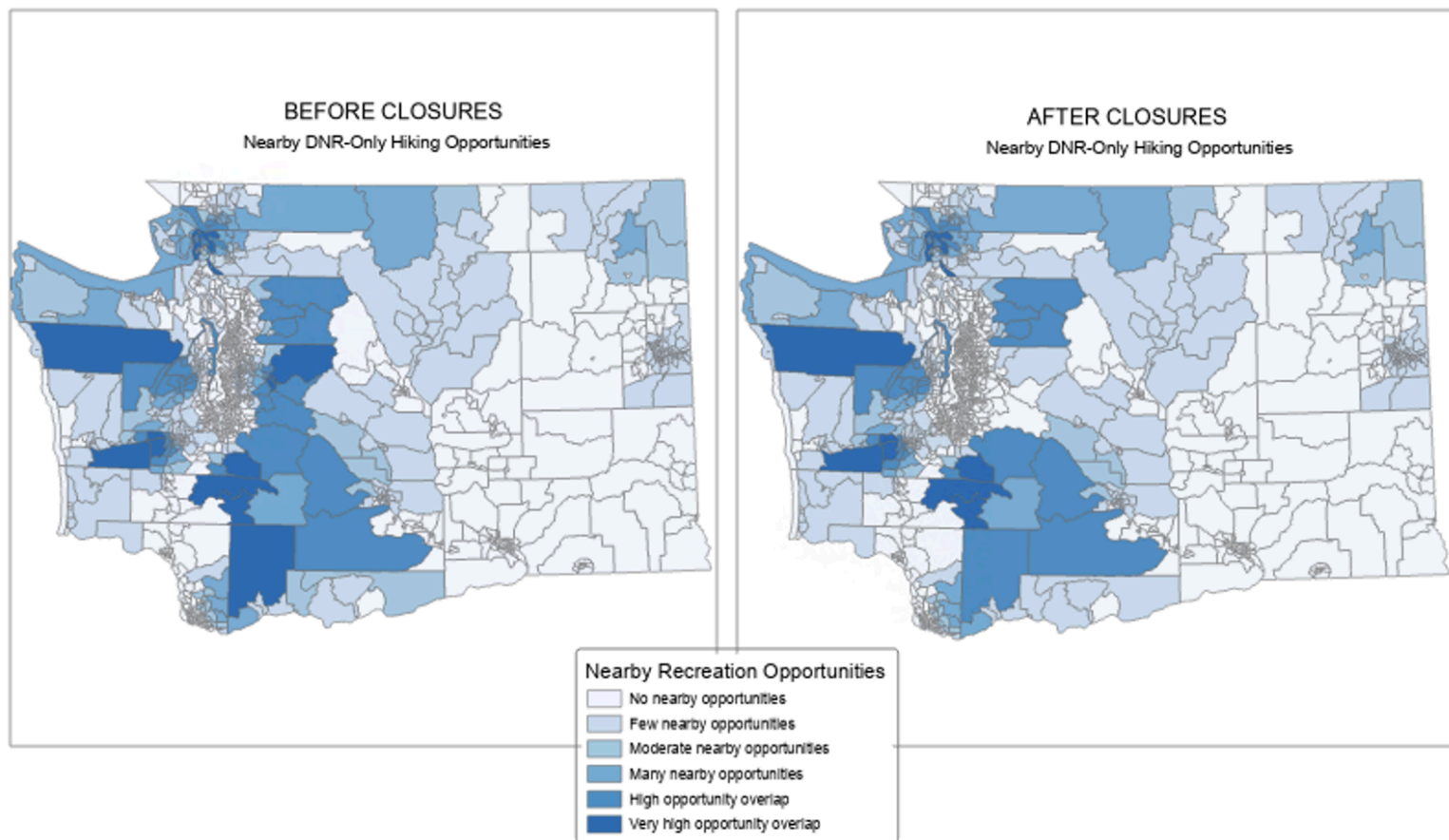


Figure 9. Comparison of tract-level modeled accessibility to nearby DNR-only hiking sites before and after DNR recreation closures. The post-closure scenario shows reduced modeled proximity to hiking opportunities across several regions, highlighting spatial differences in recreation access after closures.

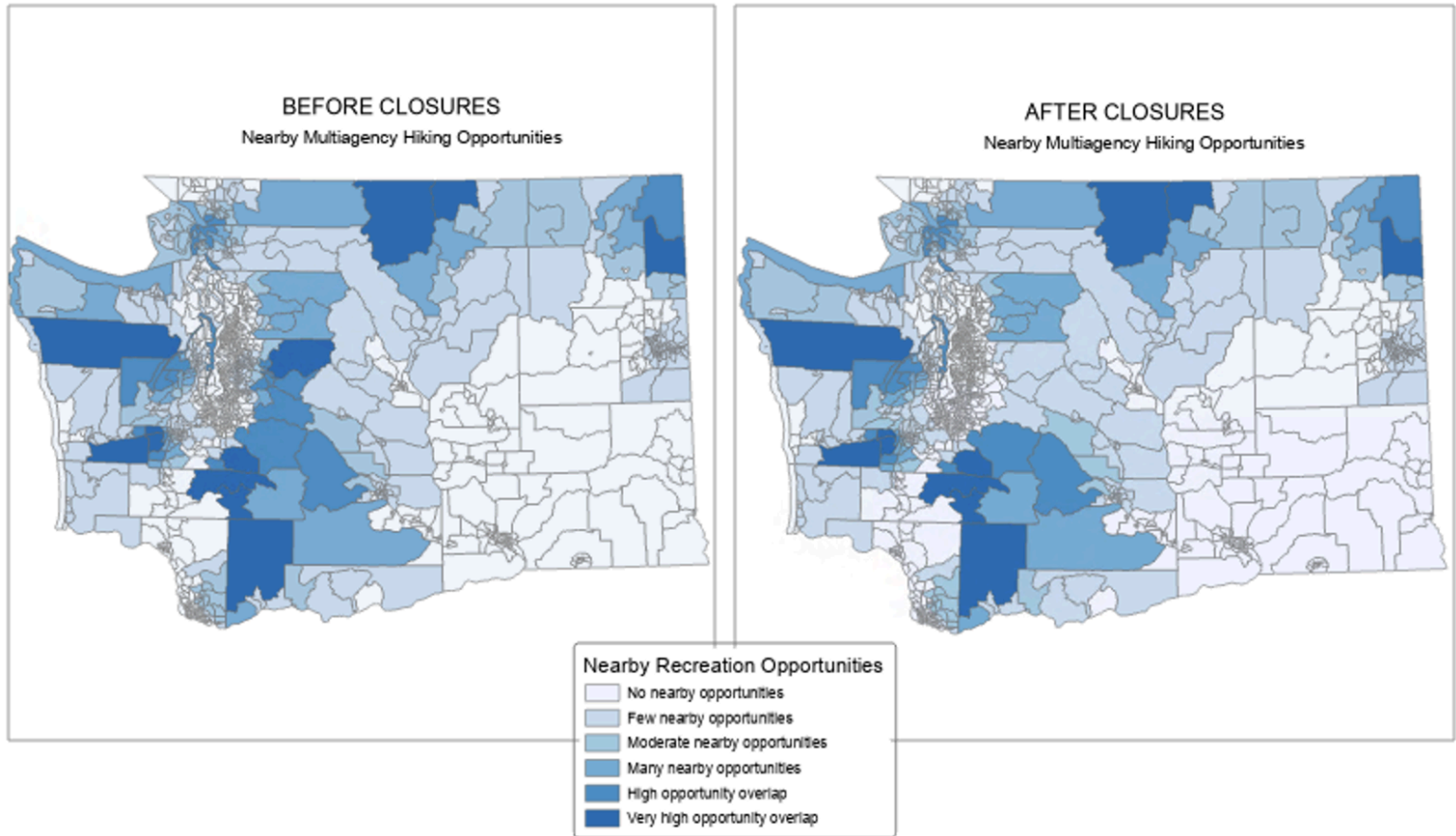


Figure 10. Comparison of tract-level modeled accessibility to nearby multiagency hiking opportunities before and after DNR recreation closures. While overlapping recreation systems help maintain accessibility in some regions, several census tracts experience reduced modeled proximity to hiking opportunities after closures.

Demographic patterns were also very similar across both scenarios. Median household income within accessible tracts dropped from about \$100,000 to around \$85,000, while the Asian population decreased from about 7.5% to 2.8%. Figures 9 and 10 also show that several areas continued to experience reduced nearby hiking opportunities even after adding recreation sites from another agency. These findings suggest that DNR appeared to function as the primary nearby hiking provider for many affected communities and that losses in hiking opportunities were not substantially offset through broader agency networks.

5.2.4 Multiagency Equity Snapshot After Closures

To evaluate remaining recreation opportunities after accounting for all available agency substitution, a final equity snapshot combined remaining open DNR, USFS, and Washington State Parks recreation sites using a 10-mile threshold across all recreation activities.

Multiagency After Closures: Equity Comparison

	Nearby Opportunities	No nearby opportunities	Difference
Median Household Income	\$94,262	\$119,731	\$25,469
Population	4,816,512	2,999,604	-1,816,908
% non-white	33.3%	40.6%	7.3%
% Hispanic	13.2%	15.8%	2.6%
% Black	3.8%	3.3%	-0.5%
% Asian	6.9%	13.0%	6.1%
% with disability	15.0%	12.2%	-2.8%
% no vehicle	6.7%	6.2%	-0.5%
% elderly	19.8%	18.1%	-1.7%

Table 4. Nearby modeled recreation opportunities are defined as Join_Count > 0 within a 10-mile threshold. Multiagency analysis includes remaining open DNR, USFS, and Washington State Parks recreation sites after closures.

After closures, 1,133 census tracts still retained at least one nearby recreation site from any agency, while 651 tracts remained without nearby opportunities. Tracts without nearby recreation opportunities had higher proportions of non-white residents (40.6% vs. 33.3%), Hispanic residents (15.8% vs. 13.2%), and Asian residents (13.0% vs. 6.9%) compared to tracts that retained nearby opportunities. Median household income was also higher in tracts without nearby recreation opportunities (\$119,731 vs. \$94,262). This likely reflects the concentration of no-access tracts in rural eastern Washington, where lower recreation site density and higher agricultural household incomes overlap, rather than suggesting there are no equity concerns. These patterns suggest that broader recreation accessibility gaps existed even before the DNR closures and remained visible after incorporating all other agencies into the analysis.

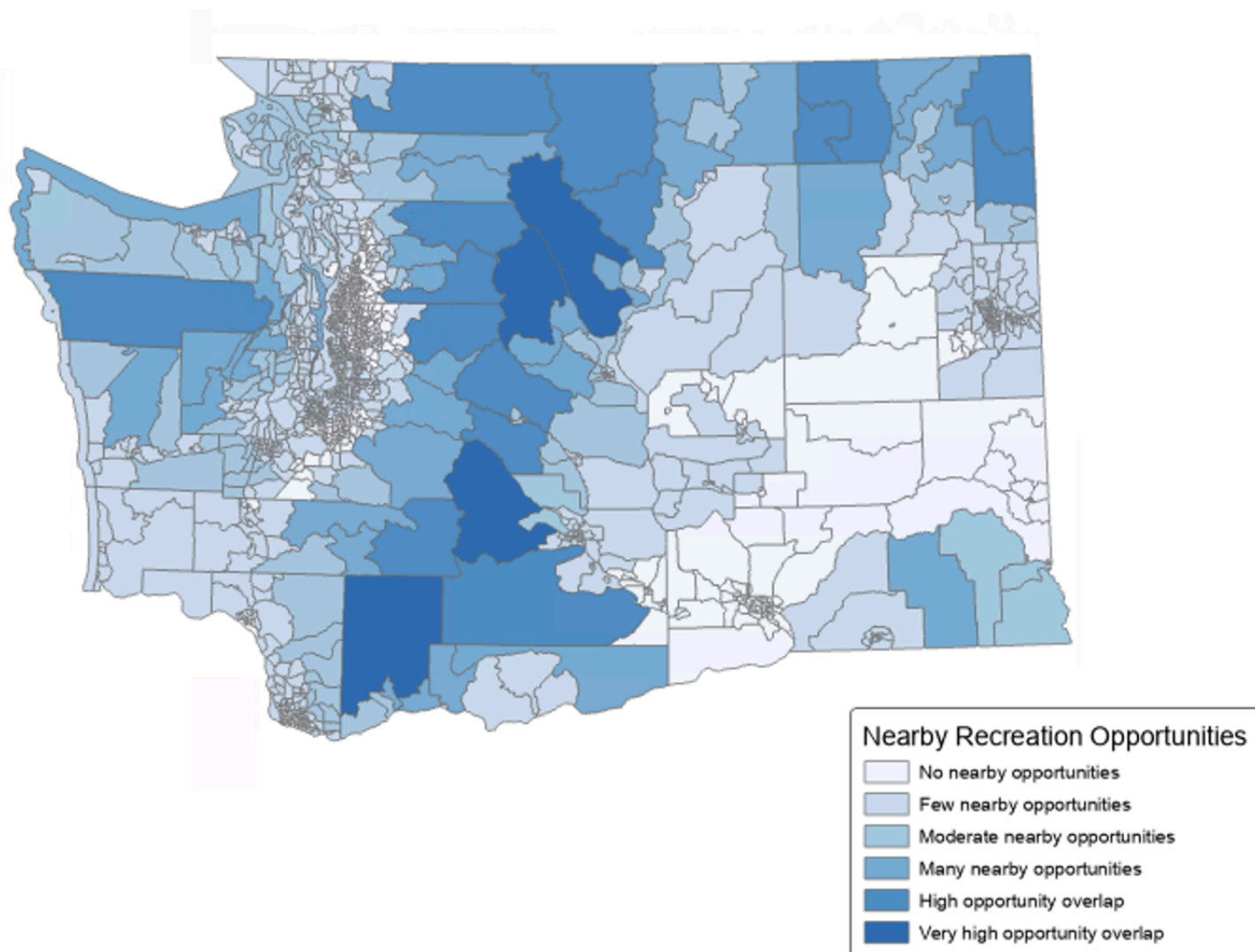


Figure 11. Multiagency modeled accessibility to recreation opportunities across Washington census tracts, incorporating open DNR, USFS, and Washington State Parks recreation sites.

Figure 11 shows that nearby access remained concentrated in western Washington and the Puget Sound region, while many eastern Washington tracts continued to have lower recreation opportunity density overall. Although multiagency recreation systems reduced some local losses, they did not eliminate broader geographic and demographic disparities in recreation proximity across the state.

Across the analyses the DNR closures reduced the number of census tracts with nearby opportunities, with hiking showing the largest overall impact. While USFS and Washington State Parks helped offset some accessibility losses for activities such as camping, they provided very limited substitution for nearby hiking sites within the 10-mile threshold. The demographic analysis also showed consistent equity patterns across multiple scenarios. Tracts without nearby access generally contained higher proportions of non-white and Hispanic residents, while

demographic shifts, particularly among Asian populations, appeared repeatedly throughout. These findings suggest that maintaining DNR hiking access would have the most meaningful impact on communities that currently lack nearby recreation alternatives.

5.3 Environmental and Health Equity Considerations

In preparation for any future service changes, this analysis combined DNR and USFS point data with Environmental and Socioeconomic data from the Washington Environmental Health Disparities Map (EHD).

The EHD index rankings were created to quantify health pressures suffered by Washington populations, pressures which stem from environmental and socioeconomic causes. For the purposes of this project, they provide a background for the visitors and users of these recreation sites.

Sites share the same rankings as their parent census tract. Sites with high rankings for EHD indicators are located in census tracts with a high ranking for that indicator. If an assumption is made that recreation sites mainly cater to the nearest local populations, then each indicator can be interpreted as a different vulnerability of the local population. This assumption may not be true, but it was assumed to be true for our health equity analysis. Under this assumption, the addition of EHD data can inform us about the living conditions from which visitors are coming. For the health equity analysis, recreation sites were conceptualized as places of escape for local populations of Washington state.

Adding the census-tract level EHD data to the individual DNR and USFS sites provides a new metric for understanding the effects of service changes on vulnerable individuals. For sites with highly vulnerable local populations, analysis can be done on the breadth of activities offered to those populations. For sites offering similar suites of activities, the burdens of the local population can be studied.

EHD data was added to both USFS and DNR point data, but our analysis is specifically about the DNR recreation sites. The USFS locations serve as alternative locations for affected activities as we evaluate the DNR site closures.

The following are some of the findings that can be derived from the joining of these datasets.

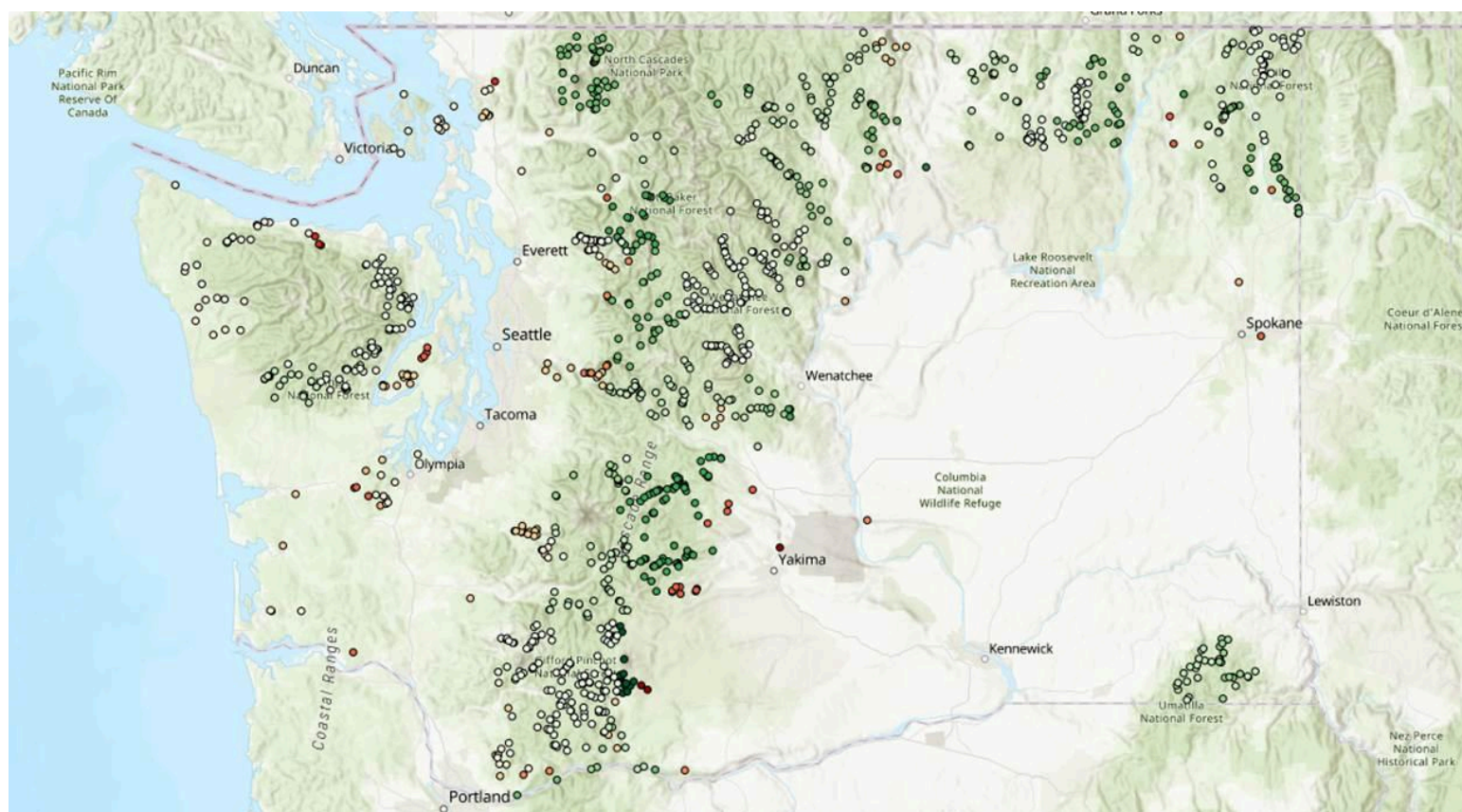


Figure 12. All DNR and USFS recreation sites in WA state, symbolized by overall EHD v2 rank.

This map shows all DNR and USFS recreation sites in Washington State, symbolized by their overall EHD V2 index rank. Lighter coloration indicates lower ranking, and darker coloration represents high ranking and higher overall health burdens. Green coloration is used for USFS sites, and Red coloration is used for DNR sites. This map shows the general overview for the level of burden experienced by the local populations of each site.

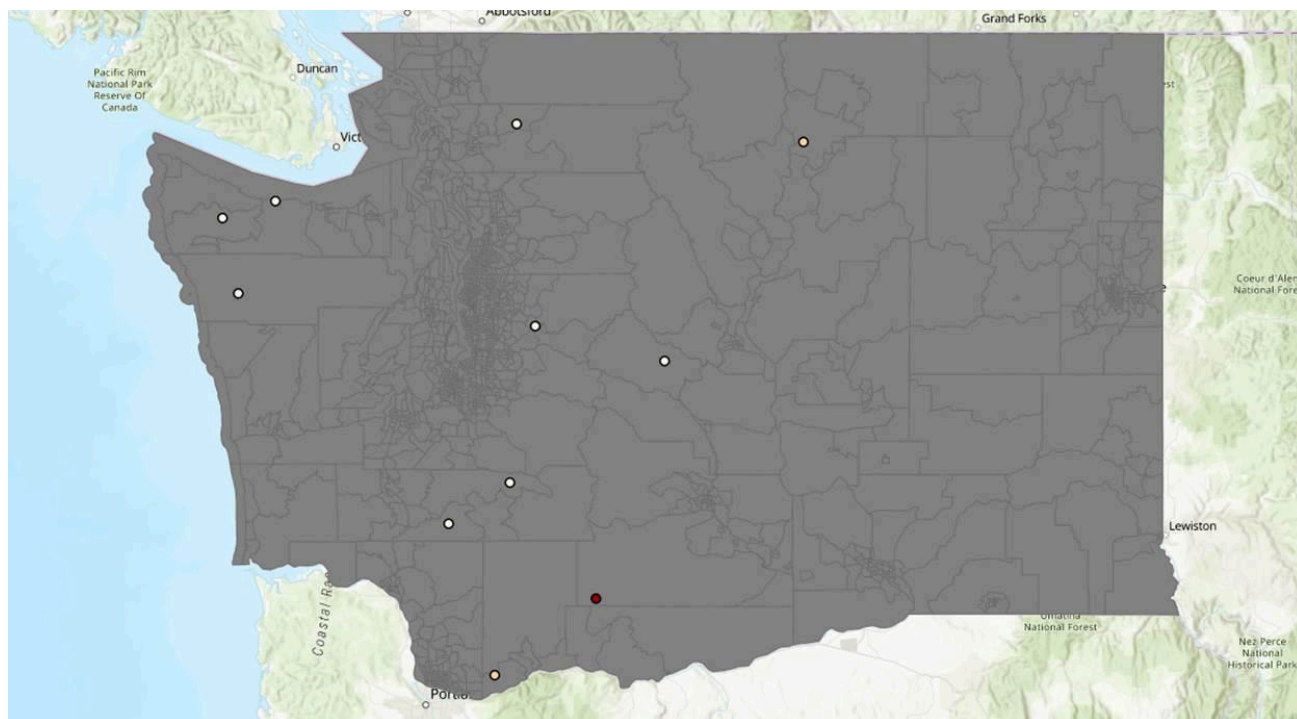


Figure 13. Finalized sites to be closed by DNR.

This map depicts the sites which will be closed by DNR for the foreseeable future. They are again symbolized by overall EHD V2 index rank. Island Camp in Yakima County ranks highest at 8, signaling the highest health and environmental burdens of these sites. The next highest ranked are Dougan Creek Campground and Rock Lakes Campground in Skamania and Okanogan Counties, respectively. The remaining locations hold a rank of 2 or less, indicating at least 80% of the other recreation sites suffer a higher level of health burden than these.

TARGET_FID	NAME	SITE_TYPE	Hiking	County10	EnvironExpos	EnvironEffect	SocioEcon	SensitivePop	EHD_V2
11	Bear Creek Campground	CG	1	Clallam	1	1	7	3	1
90	Island Camp	CG	1	Yakima	5	3	10	8	8
151	Rock Lakes Campground	CG	1	Okanogan	2	1	8	8	4
197	Upper Clearwater Campground	CG	1	Jefferson	1	1	7	3	1
213	Snoqualmie Point Trailhead	TH	1	King	4	1	1	6	2

Figure 14. Closure sites that offered Hiking.

This table includes all the closure sites which offer hiking recreation. One of them is the Snoqualmie Point Trailhead, which will remain open but with reductions to toiletry services. Examining the four main indicator categories and EHD_V2, we again see that Island Camp campground resides in a census tract with high levels of environmental and socioeconomic burden. The most severe categories are Socioeconomic and Sensitive Population, which can

represent risk of marginalization. In terms of environmental harms, it ranks five for Environmental Exposure and three for Environmental Effects. When considering the indicators of these categories (and following the assumptions made earlier), visitors to Island Camp suffer above average levels of pollution and poor air quality, and below average levels of systemic environmental hazards.

The findings at Island Camp can be compared against those for Rock Lakes Campground. In terms of Environmental Exposure and Environmental Effects, its visitors experience low levels of burden from environmental effects. However, its Socioeconomic and Sensitive Populations rankings are among the highest in Washington. So while visitors to Rock Lakes Campground are not at severe disadvantage in terms of continual environmental harms, they are potentially highly at risk from marginalization or other health considerations.

OBJECTID	TARGET_FID	NAME	SITE_TYPE	EnvironExpos	EnvironEffect	SocioEcon	SensitivePop	EHD_V2
1	2	29 Pines Campground	CG	2	1	6	4	2
2	8	Anderson Lake Campground	CG	1	1	6	3	1
3	11	Bear Creek Campground	CG	1	1	7	3	1
4	35	Chopaka Lake Campground	CG	1	1	9	3	2
5	52	Dougan Creek Campground	CG	2	4	3	7	4
6	90	Island Camp	CG	5	3	10	8	8
7	98	Les Hilde	TH	1	3	4	10	2
8	106	Lyre River Campground	CG	1	2	5	8	1
9	151	Rock Lakes Campground	CG	2	1	8	8	4
10	197	Upper Clearwater Campground	CG	1	1	7	3	1
11	204	Winston Creek Campground	CG	2	1	7	4	2
12	213	Snoqualmie Point Trailhead	TH	4	1	1	6	2

Figure 15. All closure sites with main EHD categories and overall EHD_V2 ranking

When examining the 11 full site closures and service reductions in the Snoqualmie sites, all sites were in census tracts with average to well-below average levels of environmental health burdens. The range of vulnerability based on Socioeconomic and Sensitive Population indicators, on the other hand, ranges from below average to well-above average. Combining all the indicator categorizations, all sites except for Island Camp were ranked below average for EHD_V2. With all factors considered, the local populations affected by the closures and reductions of these sites were suffering lower levels of Environmental and Health Disparity than the rest of the state of Washington.

Once again, this is under the assumption that recreation sites are most important to their local populations.

6. Summary, Conclusions, and Recommendations

Of the most impacted recreational activities we examined, most closed sites had a nearby alternative site that provided the same activity, with the exception of the Snoqualmie sites, which will not be undergoing closure but do not have any alternative sites for the activities we analyzed. There are only three closed sites that do not have an alternative site among the five most impacted activities, keeping closure impact to a minimum across Washington State. In conclusion, the DNR site closures have generally provided users with a nearby alternative site that allows people to continue enjoying the outdoor recreational activities that they lost through site closures. However, because none of the Snoqualmie sites have a suitable alternative, there needs to be a refocus of DNR priorities in providing adequate recreational service to its users.

Despite avoiding closure, due to the large quantity and popularity of Snoqualmie sites, many users will be impacted by these seasonal bathroom closures. This can worsen the experience of users on the many activities provided by these sites, especially when the time spent outdoors can be up to several hours. Without bathrooms during certain times of the year, this necessitates users to either cut their activity times short or relieve themselves in the wilderness, which can cause further complications to other users and damage to the sites. Although seasonal bathroom closures may not seem like much, they do play a notable role in shaping how users interact with each site, and can lead to negative consequences for the sites as a whole. As such, we recommend that full bathroom service be restored as quickly as possible, and if not, alternatives should be provided through transit access to nearby towns. The nearby city of North Bend, which is conveniently located along I-90, could be coordinated with to provide publicly accessible bathrooms to travelers coming to and from the Snoqualmie sites.

While many recreation activities retained nearby alternatives after closures, the analysis highlights the importance of considering both recreation type and geographic context when evaluating future service reductions. The availability of an alternative site does not necessarily mean that impacts are fully mitigated, particularly for activities and locations where substitute opportunities are limited. Future recreation planning efforts should consider not only whether alternatives exist, but also how accessible and practical those alternatives are for affected communities.

Overall, the findings demonstrate that while many recreation opportunities remained available through alternative sites, the potential impacts of closures varied by activity type, geography, and community characteristics. This project provides DNR with a framework for evaluating how future recreation system changes may affect communities across Washington State. By combining alternative site analysis, proximity-based recreation opportunity modeling, and demographic and environmental equity assessments, the project offers a more comprehensive understanding of how recreation opportunities are distributed and how potential impacts may vary across different regions and populations.

7. References

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8. Appendix

Appendix A

Emily Pitman Handy, Recreation Data Manager & GIS Analyst (Washington Department of Natural Resources) — (564) 233-8689

Appendix B

Appendix B.1 ACS Variables Used in the Analysis

Source: U.S. Census Bureau American Community Survey (ACS) 5-Year Estimates.

Variable	Description
Median Household Income	Median household income estimate for each census tract
Total Population	Total population within each census tract
Percent Non-White	Percentage of residents identifying as non-white
Percent Hispanic	Percentage of residents identifying as Hispanic or Latino
Percent Black	Percentage of residents identifying as Black or African American
Percent Asian	Percentage of residents identifying as Asian
Percent with Disability	Percentage of residents reporting a disability
Percent No Vehicle	Percentage of households without vehicle access
Percent Elderly	Percentage of residents age 65 and older

Example of ACS variable definitions used in demographic analysis.

Appendix B.2 Environmental Health Disparities



Washington Environmental Health Disparities Map

Cumulative Impacts of Environmental Health Risk Factors Across Communities of Washington State

DOH 311-011 July 2022
Updated July 2022
Version 2.0



Appendix B.3 DNR Closure Documentation

2026 Recreation Impacts Due to Budget Cuts

RECREATION SITE	REGION, CLOSEST TOWN	LEVEL OF IMPACT	EFFECTIVE DATES/NOTES
29 Pines Campground	SE Region, Cle Elum	Late opening	Campground will open July 2026 with reduced number of sites due to storm damage
Anderson Lake Campground	SPS Region, Elbe	Full closure	Will not reopen for 2026
Bear Creek Campground	OLY Region, Forks	Seasonal closure	Effective 9/15/26 to 4/15/27
Dougan Creek	PC Region, Washougal	Reduced season	Campground will open 6/1/2026; close 10/1/2026 to 6/1/2027
Harry Osborne State Forest	NW Region, Sedro-Woolley	Seasonal closure	Effective 12/1/2026 to 2/28/27
Island Camp	SE Region, Glenwood	Closed to overnight camping	Cabin and day use remain open. Overnight camping closed immediately.
Lyre River Campground	OLY Region, Port Angeles	Seasonal closure	Effective 9/15/26 to 4/15/27
Rock Lakes Campground	NE Region, Conconully	Full closure	Will not reopen for 2026
Snoqualmie Area Sites	SPS Region, North Bend/Snoqualmie	Seasonal restroom closures	Minimal bathroom cleaning services; locked toilet facilities from 10/1/2026 to 4/30/2027
Upper Clearwater Campground	OLY Region, Forks	Full closure	Campground will close for the year on 6/1/2026
Winston Creek Campground	PC Region, Mossyrock	Reduced season	Campground will open 6/1/2026; close seasonally from 10/1/2026 to 6/1/2027 (also open in December for hunting season)

Appendix C

Appendix C.1 Edwin Bai's Resume

Edwin Bai

626-234-9677 | ebai@uw.edu | [Los Angeles Journal Page](#) | [Sasians Website](#) | [LinkedIn](#)

EDUCATION

University of Washington - GPA: 3.98 Seattle, WA
B.A. in GIS, Sociology, Political Science; Minors in Urban Design, Labor Studies, Data Science Sep. 2023 – June 2026
Primary Field of Interest: Urban Planning **Secondary Field of Interest:** Network Analysis

RELEVANT EXPERIENCE AND PROJECTS

GIS Technician January 2025 – Present
BuildSD San Diego, CA

- Visualize all active and planned construction projects in Central San Diego using a 3D Model in ArcGIS Pro with LiDAR data from the USGS, 8 square miles visualized and uploaded publicly to ArcGIS Online
- Classify LiDAR point clouds into ground, vegetation, building, and noise, using the building point clouds to create a polygon layer of the building footprints, later used to create 3D buildings using LAS Building Multipatch tool
- Perform manual and automated quality control of polygon footprint layer using Eliminate Polygon Part, Regularize Building Footprints, and Edit Vertices tools to remove gaps in polygons and establish right angles in buildings
- Manually create new buildings in 3D layer to update 3D Scene with new planned and active construction projects, achieving the desired structure shape using real-world photographs and construction diagrams as reference

Student Assistant June 2025 – Present
Evictorbrook Seattle, WA

- Utilize Python and BeautifulSoup to scrape html table data from the King County Superior Court Clerk's Office regarding corporate evictions throughout the county - **750** eviction cases
- Use Python to perform data cleaning and data manipulation processes with Pandas, including sorting cases by building and company, and geocode coordinates with addresses as input using geopy.geocoders
- Utilize ArcGIS online to map the distribution of evictions by building in King County, with the intent to highlight corporations that are particularly egregious with tenant evictions - **52** buildings mapped

Research Intern June 2025 - Present
University of Washington eScience Institute Seattle, WA

- Use python and urllib.request to bulk download historical Black newspapers from the US Library of Congress website by automatically generating urls to the pdf file, bypassing the Library of Congress api limit
- Geocode addresses into coordinates using python, pandas, and google geocoding api
- Use ArcGIS Storymaps to organize a map presentation of the distribution of Black theaters over time in the US, using before and after images to showcase how individual cities have changed across time

Assistant Managing Editor May 2023 – Present
Los Angeles Journal Los Angeles, CA

- Research-writing on a variety of sociological and political topics written in a straightforward, educational format for the layman reader, edited and published using **WordPress** - **8** total publications
- Explore racial health disparities in the US, wrote a paper on how residential segregation and institutionalized racism have led to Black Americans experiencing far worse health outcomes than White Americans on average
- Interview and document experiences of Pro-Palestine activists organizing hunger strikes on California State University campuses

CERTIFICATES AND AWARDS

University of California, San Diego - CEQA Practice Certificate, Geographic Information Systems Certificate
 University of Washington Department of Geography - 2025 Gerald W. Halmo Geography Scholar

SUMMARY OF QUALIFICATIONS

Adept with ArcGIS tools and applications, including ArcGIS Pro, Online, and Storymaps. Able to conduct spatial analysis, network analysis, site suitability analysis, supervised/unsupervised classification, raster processing, LiDAR processing, and 3D modeling in ArcGIS Pro. Comfortable with creating public-facing visualizations using layouts in ArcGIS Pro. Able to upload, organize, and use Map Viewer to visualize and symbolize spatial data. Experience with ArcGIS Storymaps to convey information regarding the history of Black community and culture throughout US History through the documentation of Black cinema spaces. Created interactive maps that allowed for time-based analysis and interactive map points that provided additional information on each cinema to users.

Appendix C.2 Johann Ravindranathan's Resume

Johann Ravindranathan

jravin2@uw.edu

Student at University of Washington majoring in Geography: Data Science. Technical skills include GIS mapping, developing geospatial visuals with QGIS & ArcGIS Pro, as well as data analysis with R & Python. Experienced with Java and statistical methods. Beginner familiarity with Machine Learning.

Education

B.A. in Geography with Data Science, University of Washington, Seattle – June 2026

Skills

GIS Mapping

- Projections, Raster, Vector, Symbology, Hot Spot Analysis, Network Analysis, OD Cost Matrix, Space Time Cube

Data Analysis

- Python / R
- Pandas, GeoPandas, GeoJSON, GDAL, OGR, Enterprise GIS, Web Scraping, API data collection

Statistics

- R
- Hypothesis testing, ANOVA, Linear / Multiple Regression, Model Building + Validation

Machine Learning

- Python
- NumPy, Scikit Learn
- Regression, Classification, Clustering, Retrieval, Deep Learning

Website Development

- Python / Markdown / HTML / CSS / JavaScript

Previous Projects

[Washington EV Charger Mapping](#)

- Final project for Web GIS class
- Exploratory assignment on creating a GitHub pages – hosted website with interactive mapping features
- Collected data on EV charging infrastructure in WA state

- Inputted data into several maps to display concentrations and distances from charging networks

Mock study of Effects of Transportation on Daily Lives of UW Students

- Final project for Explanation and Understanding Geography class
- Created design for a potential research study into usage of public / private transportation by UW students, and effects on daily life outside of class
- Report included Literature Review, proposed Methodology for study, Intentions, Limitations, required Resources, Budget, and proposed benefit of study

Statistical study into Effect of Various Parameters on Ungulate Density

- Final project for Statistical Inference in Applied Research class
- Given a provided data source with a response variable and several predictor variables, conducted rigorous model testing to determine ideal model for estimating regional Ungulate densities
- Used Categorical and Continuous variables in Linear Models, applying transformations and data cleaning to achieve best results
- Final submission included detailed report of my work, and in-person presentation

Primary Field of Interest

Population movements in Washington state, such as experiences with public and private transportation and effects on daily life.

Secondary Field of Interest

Natural environments and resources, such as animal populations, ecosystem health, resource extraction and reserves, human habitation trends.

Appendix C.3 Max Marchand's Resume

Max Marchand

360-742-4519 | mfm23@uw.edu | [linkedin.com/in/max-marchand](https://www.linkedin.com/in/max-marchand)

EDUCATION

University of Washington Seattle, WA
Bachelor of Arts in Geographic Information Systems and Data Science Sep. 2022 – June 2026
 • **Minors:** Business & Urban Design and Planning
 • **Honors & Leadership:** Dean's List — Club Soccer Captain — Film Club Lead Critic

EXPERIENCE

Geographic Information Systems Intern June 2025 – Present
King County Department of Natural Resources and Parks Seattle, WA
 • Created feature class dataset containing Facility points that correspond with researched coordinates
 • Projected maintenance holes and identified positional discrepancies between the current features and new layer
 • Developed methodology for identifying and projecting maintenance holes for quality control on existing data
 • Designed StoryMap presentation of the purpose, process, and findings of the conveyance system project

Dam Safety Intern June 2024 – Present
Seattle City Light Seattle, WA
 • Managed and organized all Dam Safety related Databases and Documents
 • Led the complete digitization of Dam Safety archives, ensuring easy accessibility via SharePoint
 • Utilized ArcGIS Pro and LiDAR data for potential inundation mapping
 • Developed spreadsheets tracking the location and status of both digital and physical documents
 • ESRI Certified in ArcGIS Pro - Real World GIS

Urban Planning and GIS Consultant March 2025 – June 2025
Pacific County South Bend, WA
 • Consulted on the redesign and potential urban development of Bendiksen Landing
 • Led the flood risk analysis for the site and visualized the impervious surfaces available for development
 • Analyzed water lines used for the site by georeferencing engineering drawings to project current water lines
 • Developed a parcel map to visualize ownership and acreage of the site and surrounding area
 • Presented maps and comprehensive plan to Pacific County

Rec Clubs Program Coordinator October 2023 – June 2025
University of Washington Seattle, WA
 • Oversaw budget details for all recreational sports clubs across campus.
 • Managed club files and redesigned SharePoint for improved accessibility and efficiency
 • Used Fusion and Microsoft Excel to develop practice schedules for all clubs
 • Recognized as 2023/24 and 2024/25 Rec Clubs Employee of the Year

PROJECTS

Seattle Spurs | Board Member, Staff Writer, Podcast Host April 2023 – Present
 • Board Member of the Official Tottenham Hotspur Supporters Club of Seattle
 • Founding Staff Writer and Podcast Host for the Emerald and Lilywhite blog
 • Record episodes twice a week analyzing the latest matches and news covering the team

Map Portfolio | Spatial Analysis, Urban GIS, Inundation Mapping January 2024 – Present
 • Collection of academic and professional GIS projects.
 • Maps of Gentrification in Seattle with supplemental thesis
 • Maps of Real Estate in Seattle around Light Rail stations with trend analysis

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL, R
GIS Software: ArcGIS Pro, QGIS, ArcGIS Online, StoryMaps
Developer Tools: Git, RStudio, Google Cloud Platform, VS Code, JupyterHub, Anaconda, IntelliJ, AutoCAD
File Management: Microsoft Excel, SharePoint, File Explorer, Adobe Acrobat, BlueBeam
Libraries: pandas, NumPy, Matplotlib

Appendix C.4 Shayla Guieb's Resume

Shayla Guieb

sguieb@uw.edu | linkedin.com/in/shaylaguieb | (425) 365-7636

Data-focused Geography student experienced in GIS, spatial analysis, and web mapping for urban systems and accessibility.

Skills

GIS & Spatial Analysis: QGIS, ArcGIS, OpenStreetMap, Spatial Joins, Web Mapping (Mapbox GL JS)

Programming, Web Development, & Data: JavaScript, HTML, CSS, Python, R, Pandas, Data Cleaning

Tools: Git, GitHub

Projects

Seattle Transit Accessibility & Equity Analysis – GIS Dashboard *QGIS, JavaScript, HTML, CSS, GitHub*

- Conducted buffer and spatial join analysis to evaluate transit access across census tracts
- Integrated transit, route frequency, and demographic datasets to analyze mobility equity
- Built interactive maps and a dashboard to communicate spatial disparities in transit access

Gentrification and Housing Trends in Seattle – Story Map *ArcGIS, Carto, Spatial Data Analysis*

- Integrated 5+ datasets (HUD, Census, Seattle Open Data) across 20+ ZIP codes (2000–2024)
- Analyzed relationships between rent, income, and eviction patterns using spatial joins and temporal comparison
- Designed 5 interactive maps (choropleth + layered indicators) to highlight displacement risk

Green Lake Business Clustering – Spatial Analysis & Visualization *Python, Web Scraping, Data Visualization*

- Collected and cleaned 100+ geocoded business records for spatial analysis
- Applied clustering methods to identify patterns in commercial activity
- Visualized results using radial dendrograms and geospatial mapping

Experience

Madewell – Sales Associate Seattle, WA | June 2025 – Present

- Assisted 40+ customers per shift with personalized recommendations
- Collaborated with a team to maintain store layout and improve accessibility and flow

Molly Moon's Homemade Ice Cream – Scooper Lead Seattle, WA | 2024 – June 2025

- Led a 5-person team during high-volume shifts, coordinating workflow and responsibilities
- Delivered consistent customer service while managing fast-paced operations
- Trained new hires on systems and standards, improving team efficiency

Education

University of Washington, Seattle, WA

B.A. Geography: Data Science | Expected June 2026

- Focus on geospatial analysis, spatial visualization, and accessibility-focused digital tools

Fields of Interest

Primary: GIS & Spatial Data Analysis (Transit Accessibility, Urban Systems)

Secondary: Data Science & Visualization (Web Mapping, Interactive Dashboards)